

Mathematics, 2018 EC, entrance exam, version 2

Grade 12 · Mathematics · 2018 E.C.

Time allowed: 120 minutes

Questions: 56

Total marks: 56

Name _____ Grade _____ ID No. _____ Date _____

1. What is the 10th term of Prof. Mulatu's sequence ? (1)

A. 309

B. 118

C. 191

D. 89

2. What is the area (in square unit) of the region bounded by $y=0$ and the graph of $f(x)=4-x^2$ on the interval $[-2,2]$? (1)

A. $2/3$

B. $22/3$

C. 8

D. $32/3$

3. Let f be defined on $[a, b]$. Then which of the following is NOT true? (1)

A. $\int_a^b f(x)dx = \int_a^c f(x)dx - \int^cb f(x)dx$ for any c in (a,b)

B. $\int^{cc} f(x)dx = 0$ for any c in (a,b)

C. $\int_a^b f(x)dx = -\int^ba f(x)dx$

D. $\int_a^b k f(x)dx = k \int_a^b f(x)dx$ for any constant k

4. The sides of an equilateral triangle are increasing at the rate of 2 cm/sec. The rate at which the area increases when side is 10 cm is: (1)

A. $10 \text{ cm}^2/\text{sec}$

B. $\sqrt{3} \text{ cm}^2/\text{sec}$

C. $10\sqrt{3} \text{ cm}^2/\text{sec}$

D. $10/3 \text{ cm}^2/\text{sec}$

5. A balloon which always remains spherical is being inflated by pumping in 10 cm^3 of gas per second. What is the rate at which the radius is increasing when the radius is 15 cm? (1)

A. $1/(30\pi) \text{ cm/sec}$

B. $1/(90\pi) \text{ cm/sec}$

C. $1/\pi \text{ cm/sec}$

D. $1/(9\pi) \text{ cm/sec}$

6. Suppose Target Population $N = 54000$ vouchers and sample size $n = 6000$; following the systematic sampling approach if 4th unit is selected as the first unit in the sample then the next unit selected in the sample would be: (1)

A. 8th

B. 10th

C. 12th

D. 13th

7. The standard deviation and arithmetic mean of aggregate marks obtained by three groups of students X, Y, Z are as follows: X: mean=53, sd=211; Y: mean=83, sd=19; Z: mean=50, sd=10. The more variable group is: (1)

A. X

B. Y

C. Z

D. X&Z

8. When the variability of two series is to be compared, which of the following should be used? (1)

A. Absolute measure of average

B. Absolute measure of dispersion

C. Relative measure of dispersion

D. Relative measure of average

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9. A rectangular page contains 24 square inches of print. The margins: top and bottom 1.5 inch, left and right 1 inch. (1)
What dimensions of the page produce the smallest area?
- A. 6 by 4
B. 8 by 6
C. 6 by 9
D. 10 by 8
-
10. If there are many extreme scores in all examinations, the dispersion is (1)
- A. Large
B. Small
C. Normal
D. Symmetric
-
11. If the values of variable x are -4, -20, -30, -44 and -36, then the value of range will be (1)
- A. -48
B. 40
C. -40
D. 48
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12. A particle moves along the x -axis with acceleration $a(t)=12t^2+6t$ for $t \geq 0$. If $v(1)=5$ and $s(0)=2$, then what is (1)
position at $t=2$ sec?
- A. 26 m
B. 12 m
C. 24 m
D. 22 m
-
13. If p th term of an arithmetic progression is q and its $(p+q)$ th term is 0, then q th term is equal to: (1)
- A. $-p$
B. $p+q$
C. p
D. $p-q$
-
14. Which one of the following sequence is both arithmetic and geometric? (1)
- A. $-1, 1, -1, 1, \dots$
B. $1/3, 1/3, 1/3, 1/3, \dots$
C. $0, 2, 2, 2, \dots$
D. $25, 5, 1/5, 1/25, \dots$
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15. A ball is dropped from a height of 64 meters. Each time it bounces, it reaches half the previous height. Total (1)
distance traveled before coming to rest?
- A. 128 m
B. 192 m
C. 256 m
D. 320 m
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16. If 7th and 21st terms of arithmetic sequence are 24 and 52 respectively, then sum of first 5 terms is: (1)
- A. 65
B. 80
C. 82
D. 90
-
17. If the sequence defined recursively by $a_1=1$, $a_2=-1$ and $a_n = a_{n-1} + 2a_{n-2} + n$ for $n \geq 3$, then seventh term is: (1)
- A. 82
B. 75
C. 120
D. 37
-
18. If the sum of first n terms of arithmetic progression is $2n+3n^2$, then a_n is equal to (1)
- A. $4n+3$
B. $6n+5$
C. $6n-1$
D. $4n-1$
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19. What is equation of normal line to the graph of $f(x)=(1-x^3)\sqrt{x+5}$ at $(-1,4)$? (1)
- A. $2y+11x=-3$
B. $2y-11x=19$
C. $11y-2x=46$
D. $11y+2x=42$

20. If $f(x)=x^3+ax^2+bx-2$ has relative minimum value of 6 at $x=1$, then equation of tangent line at $x=-1$ is: (1)
- A. $y-40x=10$ B. $y=40x-10$
C. $y-40x=70$ D. $y=40x-70$
21. In which case is the interquartile range preferred over the range? (1)
- A. When data is binary
B. When focusing on extreme values
C. When analyzing the middle 50% of the data
D. For data set with identical values
22. In case of positively skewed distribution, the extreme values lie in the: (1)
- A. Middle B. Left tail
C. Right tail D. Anywhere
23. The output of 20 workers: 55,65,62,60,74,75,70,70,72,67,78,79,80,68,54,56,63,69,63,71. Then coefficient of range is: (1)
- A. 0.29 B. 0.19
C. 0.49 D. 0.39
24. The following data: 50,69,20,33,53,39,40,65,59. The mean deviation about the median is: (1)
- A. 9 B. 10.5
C. 12.67 D. 14.76
25. A smaller value of coefficient of variation indicates: (1)
- A. Higher variability B. More skewness
C. Higher consistency D. More central tendency
26. Bowley's coefficient of skewness lies between: (1)
- A. 0 and 1 B. -1 and 1
C. -1 and 0 D. -2 and 2
27. The lower and upper quartiles are 80 and 120, median 100. The shape of distribution is: (1)
- A. Positively skewed B. Negatively skewed
C. Symmetrical D. Elongated to the right
28. The sum of the deviations of all data points from the mean is always: (1)
- A. Positive B. Zero
C. Negative D. Equal to range
29. What is the sum of all integers between 20 and 201 that are divisible by 2 or 5? (1)
- A. 16,110 B. 12,970
C. 16,150 D. 11,970
30. Which one of the following is true about $f(x)=x^4 - (4/3)x^3$? (1)
- A. f is decreasing on $[1,\infty)$
B. The function has no relative extreme at $x=0$
C. f is increasing on $(-\infty,1]$
D. $f(1)$ is local maximum

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31. A particle position $s(t)=t^4-3t+5$. Which one is true? (1)
- A. $v(t)=t^5/5 - 3t^2/2 + 5t$ B. $v(2)=4.4$ m/s
C. Particle at rest at $t=3/4$ sec D. $a(1)=12$ m/s²
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32. If $f(2)=-3$, $f'(2)=4$, $g(1)=-5$, $g'(1)=1$, and $F(x)=f(2x+2)g(1-x^2)$, then $F'(0)$ is: (1)
- A. -40 B. 0
C. -20 D. 19
-
33. If $f(x)=2x^2+5x+8$, then $[f(2+h)-f(2)]/h$ as $h \rightarrow 0$ is equal to (1)
- A. 9 B. 15
C. 13 D. 26
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34. In a geometric sequence of even number of terms, the sum of all terms is 5 times the sum of the odd terms. The common ratio is (1)
- A. 4 B. 5
C. $3/5$ D. $1/5$
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35. Let F be the anti-derivative of f . Which of the following is NOT true? (1)
- A. $\int d/dx f(x) dx = f(x)+c$ B. $\int F(x) dx = f(x)+c$
C. $F'(x)=f(x)$ D. $d/dx \int F(x) dx = F(x)$
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36. Suppose $f(x)$ defined on $[0,3]$ such that $\int_0^3 f(x)dx=0$. Which must be true? (1)
- A. $f(x)$ is identically zero B. $f(x)$ is non-negative
C. $f(x)$ must change sign D. $f(x)$ has the same sign
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37. If $f'(x)=x^2+5$ and $f(0)=-1$, then $f(x)$ is: (1)
- A. x^3+5x+1 B. x^3+5x-1
C. $1/3 x^3+5x-1$ D. $1/3 x^3+5x+1$
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38. If $f(x) = \{ x \text{ for } x < 1, x-1 \text{ for } x \geq 1 \}$, then $\int_0^2 x^2 f(x) dx$ is: (1)
- A. 1 B. $4/3$
C. $5/3$ D. $5/2$
-
39. Which one defines arithmetic sequence? (1)
- A. Ratio between consecutive terms is non-zero constant
B. Difference between consecutive terms is positive constant
C. Difference between consecutive terms is constant
D. Ratio between consecutive terms is positive constant
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40. Average rate of change of $f(x)=x\sqrt{x} - x^2 + 4x - 2$ between $x=1$ and $x=4$ is: (1)
- A. $-10/6$ B. $4/3$
C. 3 D. B?
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41. Which best describes instantaneous rate of change? (1)
- A. Slope of secant line B. Slope of tangent line at a point
C. Rate over entire period D. Same as derivative over interval

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53. Which one is anti-derivative of $f(x)=3/(x\sqrt{x})$? (1)
- A. $-6/\sqrt{x} + c$ B. $-6\sqrt{x} + c$
C. $6/\sqrt{x} + c$ D. $-3/(2\sqrt{x}) + c$
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54. Suppose $\int_8^6 f(x)dx = 21$ and $\int_1^6 f(x)dx = -12$, then $\int_1^8 f(x)dx$ is: (1)
- A. 9 B. -9
C. 33 D. -33
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55. If f and g are defined on $\mathbb{R}$ and $a,b \in \mathbb{R}$, which is necessary true? (1)
- A. If $\int f = \int g$, then $f=g$ B. If $f'=g'$, then $\int f = \int g$
C. If $f(x) \geq 3$ on $[-2,2]$, then $\int_{-2}^2 f \geq 12$ D. $\int f = \int f$
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56. Mean deviation computed from a set of data is always: (1)
- A. Negative B. Equal to standard deviation
C. More than standard deviation D. Less than standard deviation

— END OF EXAMINATION —